THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 7.8 g/cm³	Density = mass ÷ volume
2	**(b)** water	Medium in larger amount
3	**(d)** a fixed quantity (100 mL) of solvent at a given temperature	Depends on temperature
4	**(c)** warm water	Oxygen solubility decreases
5	**(c)** measuring cylinder	Graduated cylinder
6	**(b)** 65°	Angle of reflection equals angle of incidence
7	**(d)** normal	Measured from normal
8	**(c)** plane, concave and convex mirrors	All mirrors
9	**(c)** convex mirror	Like back of a spoon
10	**(c)** convex mirror	Gives a wider field of view

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ice is less dense than water (1). Its particles are farther apart (1).	1+1
12	Ice layer insulates the water below (1). Fish survive in cold weather (1).	1+1
13	Convex is thicker at centre and converges light (1). Concave is thinner at centre and diverges light (1).	1+1
14	Convex lens (1). Gives enlarged erect image when object is close (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solid–liquid: solid is solute, liquid solvent (1½). Liquid–liquid: smaller amount is solute (1½).	1½+1½
16	Left–right reversal in the image (1½). Writing appears reversed, e.g. ‘AMBULANCE’ is written reversed so it reads correctly in a rear-view mirror (1½).	1½+1½
17	Dissolved oxygen is less in warm water (1½). Pollution reduces oxygen further (1½).	1½+1½
	OR: Mass is amount of matter, kg or g (1½). Volume is space occupied, m³ or cm³ (1½).	1½+1½
18	Image is as far behind the mirror as the object is in front (1). Distance = 2 m + 2 m (1). 4 m (1).	1+1+1
	OR: Convex mirror (1½). Concave lens (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cylinder on flat surface (1). Eye level with meniscus (1). Read bottom of the curve (1). Balance zero before weighing (1). Record units (1).	1+1+1+1+1
	OR: Salt–water (1). Alcohol–water (1). Soda water (1). Brass (1). Definition of solution (1).	1+1+1+1+1
20	Light bends between media (1). Pencil activity (1). Speed changes in water (1). Depth appears less (1). Lenses use refraction (1).	1+1+1+1+1
	OR: Spectacles correct vision (1½). Magnifier uses convex lens (1½). Camera focuses image (2).	1½+1½+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1 g/cm³	1000 kg/m³
2	**(c)** arrange to take up more space	Expansion on freezing
3	**(a)** greater density	Dissolved salt increases density
4	**(a)** soda water	CO_2 dissolved in water
5	**(d)** adding more solvent	More capacity
6	**(d)** inverted	Depends on distance
7	**(d)** converge	Converging lens
8	**(d)** refraction	Speed changes
9	**(a)** Concave mirror – converging	Concave mirror converges
10	**(d)** also reflect light (diffuse reflection)	Light scatters in many directions

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Density = 54 ÷ 20 (1). 2.7 g/cm³ (1).	1+1
12	Volume = mass ÷ density (1). 250 cm³ (1).	1+1
13	Bending of light when it passes from one medium to another (1). Pencil appears bent in water (1).	1+1
14	Light bends while going from water to air (1). Eye sees it from different direction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume = 2000 cm³ (1). Mass = density × volume (1). 2000 g = 2 kg (1).	1+1+1
16	Rough surface has irregularities (1). Normals point in different directions (1). Reflected rays scatter, giving diffuse reflection (1).	1+1+1
17	Oil has a lower density than water (1½). Hence oil rests above water (1½).	1½+1½
	OR: Stirring makes dissolving faster (1½). Solubility depends on temperature, not on stirring (1½).	1½+1½
18	Plane: same-size erect face (1). Convex: smaller face (1). Concave: enlarged face (1).	1+1+1
	OR: They need an enlarged image (1½). Concave mirror gives erect magnified image at short distance (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Density less than water floats (1). Greater sinks (1). Wood example (1). Iron example (1). Ice example (1).	1+1+1+1+1
	OR: Add solute till no more dissolves (1). Add more solvent (2). Raise temperature (2).	1+2+2
20	(i) 30° each (1). (ii) 60° (1). (iii) 60° (1). (iv) Angle becomes 0° (1). Ray retraces its path (1).	1+1+1+1+1
	OR: Reflectors direct sunlight (1). Light converges at a point (1). Pot kept at that point gets heat (1). Concave mirror converges (1). Saves fuel and electricity (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 100 g	Mass = density × volume
2	**(c)** present in smaller amount	Rule for liquid mixtures
3	**(b)** increases	Warm water dissolves more sugar
4	**(d)** matter in an object	Measured in kg or g
5	**(b)** meniscus	Read at the eye level
6	**(b)** 50°	Angle with surface = 90° − angle of reflection
7	**(b)** equal to angle of reflection	First law
8	**(d)** erect and the same size as the object	Virtual, laterally inverted
9	**(c)** converge	Converging mirror
10	**(d)** give a larger view area	Wide view

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Generally decreases with rise in temperature (1). Because substances expand (1).	1+1
12	Salt in water (1). Alcohol in water (1).	1+1
13	Erect (1). Diminished (1).	1+1
14	Concave mirror (1). Depends on object distance (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solid solute dispersed in a solid solvent (1½). Brass – zinc in copper (1½).	1½+1½
16	Concave lens (1½). It diverges rays so image falls properly on the retina (1½).	1½+1½
17	Volume = 8 cm³ (1). Density = 24 ÷ 8 (1). 3 g/cm³ (1).	1+1+1
	OR: Boat's shape has large volume with air (1½). Average density is less than water (1½).	1½+1½
18	No (1). Plane mirror forms a virtual image (1). It cannot be caught on a screen (1).	1+1+1
	OR: Angle of reflection = 50° (1½). Total angle = 100° (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = 1 g/mL (1½). B = 0.8 g/mL (1½). C = 1.2 g/mL (1). B floats (1).	1½+1½+1+1
	OR: Gas dissolved under pressure (1). Opening releases pressure (1). Heat reduces solubility (1). Gas escapes as bubbles (1). Drink goes flat (1).	1+1+1+1+1
20	Concave mirrors give magnified erect images (2). Plane gives same-size image (1). Convex gives smaller image (1). Hence concave is chosen (1).	2+1+1+1
	OR: Convex mirror gives wide view (1). Image is erect (1). Image is smaller (1). Distances look farther (1). Hence objects are closer than they appear (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** less than water	Lighter per volume
2	**(a)** it insulates water below for aquatic life	Fish survive in cold weather
3	**(a)** add salt to the water	Increase water density
4	**(d)** zinc is mixed uniformly in copper	Alloy
5	**(c)** 1 g	Density 1 g/cm³
6	**(c)** concave	Converge light for heat
7	**(b)** diverge	Diverging lens
8	**(d)** refraction of light	Light bends at the surface
9	**(a)** Convex lens – converging	Convex lens converges
10	**(d)** reflects sunlight	Not a luminous body

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gas is less soluble at higher temperature (1). Gas escapes (1).	1+1
12	Balance measures mass (1). Cylinder measures volume of liquids (1).	1+1
13	It reflects sunlight (1). Reflected light reaches Earth (1).	1+1
14	Regular reflection from smooth surface (1). Diffuse reflection from rough surface (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Each substance has a characteristic density (1½). Measured density can be compared with known values (1½).	1½+1½
16	Normal tilts with the mirror (1½). Angle of incidence changes while laws still hold (1½).	1½+1½
17	They mix uniformly (1½). Smaller quantity is solute and larger is solvent (1½).	1½+1½
	OR: Solubility reduces (1½). Excess solute crystallises out (1½).	1½+1½
18	Area visible in a mirror (1). Convex mirror gives the largest (1). Used in shops or vehicles for safety (1).	1+1+1
	OR: Mirror reflects; lens refracts (1). Mirror converges; lens diverges (1). Mirror can form real images; lens forms virtual images (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solubility decreases (1). Excess solute separates (1). Crystals form (1). Example – sugar or salt crystals (1). Slow cooling gives bigger crystals (1).	1+1+1+1+1
	OR: Sea water has salts (1). Blood carries dissolved substances (1). Oxygen dissolved for fish (1). Alloys in industry (1). Medicines as solutions (1).	1+1+1+1+1
20	Convex enlarges text at a short distance (2). Concave always diminishes (1). Flat glass shows same size (1). Conclusions on lens types (1).	2+1+1+1
	OR: Regular reflection by smooth surfaces (1). Diffuse reflection by rough surfaces (1). Light scatters in many directions (1). Reflected light enters our eyes (1). So we see the Moon and other objects (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.